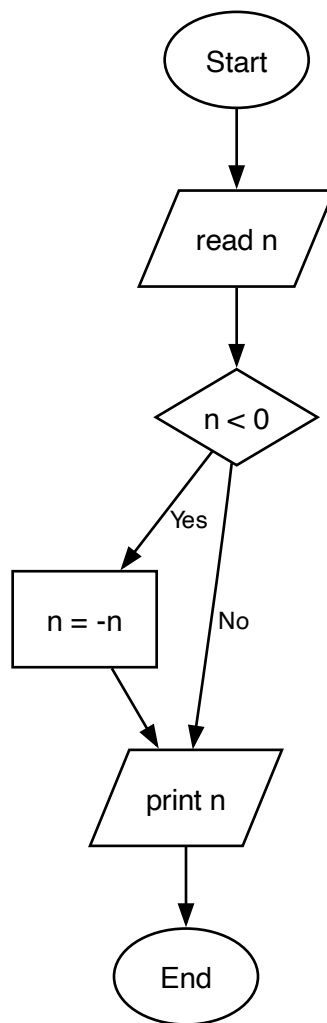


Answer key

**Problem 1.**



(i) For each input below, follow the flowchart and write the output.

| input $n$ | output |
|-----------|--------|
| 7         | 7      |
| -4        | 4      |
| 0         | 0      |
| -19       | 19     |

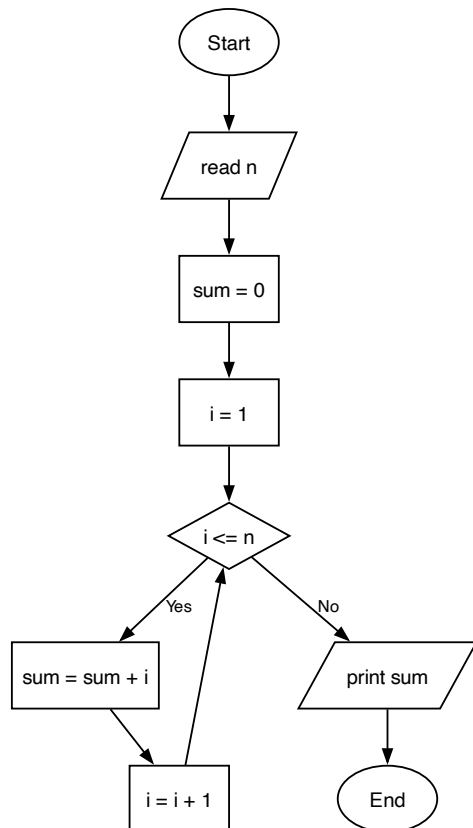
(ii) Can you tell what this algorithm is doing?

**Answer:** Absolute value

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## Problem 2.

Trace this algorithm for  $n = 4$ , and complete the trace table.



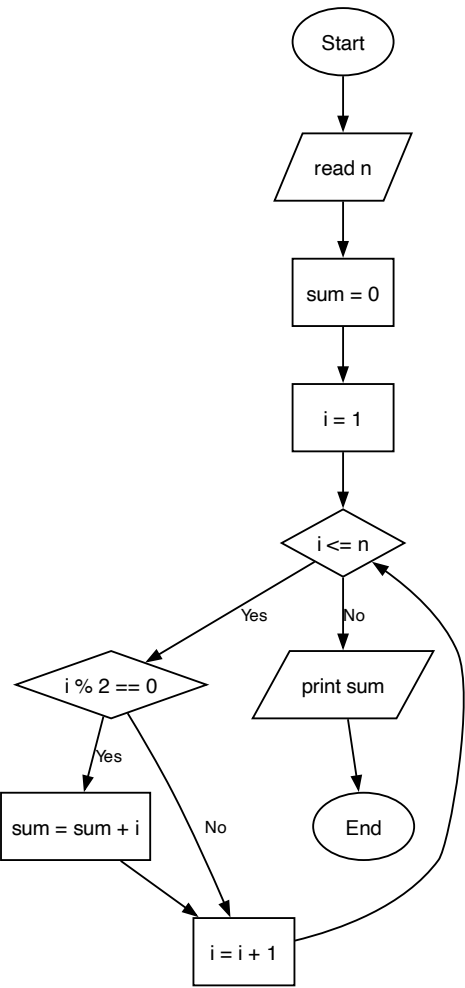
| n | sum | i | output |
|---|-----|---|--------|
| 4 |     |   |        |
| 4 | 0   |   |        |
| 4 | 0   | 1 |        |
| 4 | 1   | 1 |        |
| 4 | 1   | 2 |        |
| 4 | 3   | 2 |        |
| 4 | 3   | 3 |        |
| 4 | 6   | 3 |        |
| 4 | 6   | 4 |        |
| 4 | 10  | 4 |        |
| 4 | 10  | 5 |        |
| 4 | 10  | 5 | 10     |

**Output:** 10

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**Problem 3.**

Trace this algorithm for  $n = 5$ , and complete the trace table.

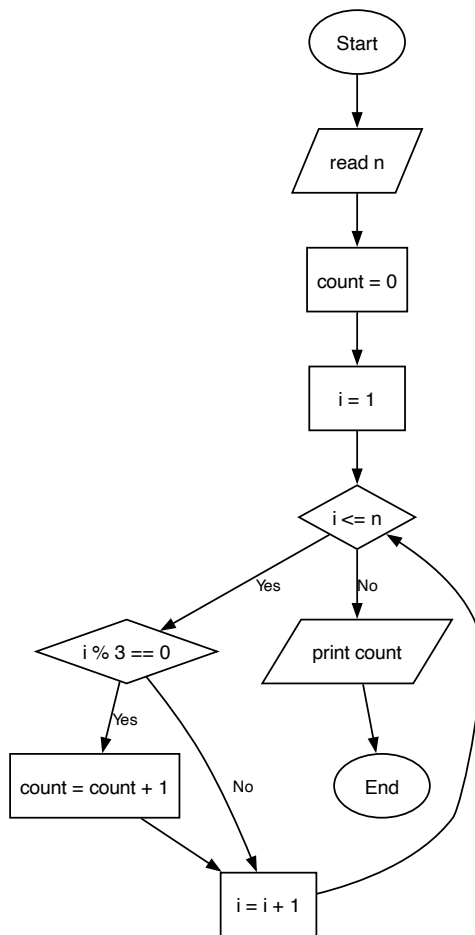


| n | sum | i | output |
|---|-----|---|--------|
| 5 |     |   |        |
| 5 | 0   |   |        |
| 5 | 0   | 1 |        |
| 5 | 0   | 2 |        |
| 5 | 2   | 2 |        |
| 5 | 2   | 3 |        |
| 5 | 2   | 4 |        |
| 5 | 6   | 4 |        |
| 5 | 6   | 5 |        |
| 5 | 6   | 6 |        |
| 5 | 6   | 6 | 6      |

**Output:** 6

#### Problem 4.

Trace this algorithm for  $n = 7$ , and complete the trace table.



| n | count | i | output |
|---|-------|---|--------|
| 7 |       |   |        |
| 7 | 0     |   |        |
| 7 | 0     | 1 |        |
| 7 | 0     | 2 |        |
| 7 | 0     | 3 |        |
| 7 | 1     | 3 |        |
| 7 | 1     | 4 |        |
| 7 | 1     | 5 |        |
| 7 | 1     | 6 |        |
| 7 | 2     | 6 |        |
| 7 | 2     | 7 |        |
| 7 | 2     | 8 |        |
| 7 | 2     | 8 | 2      |

**Output:** 2

### Problem 5.

Trace this algorithm for  $n = 472$ , and complete the trace table.

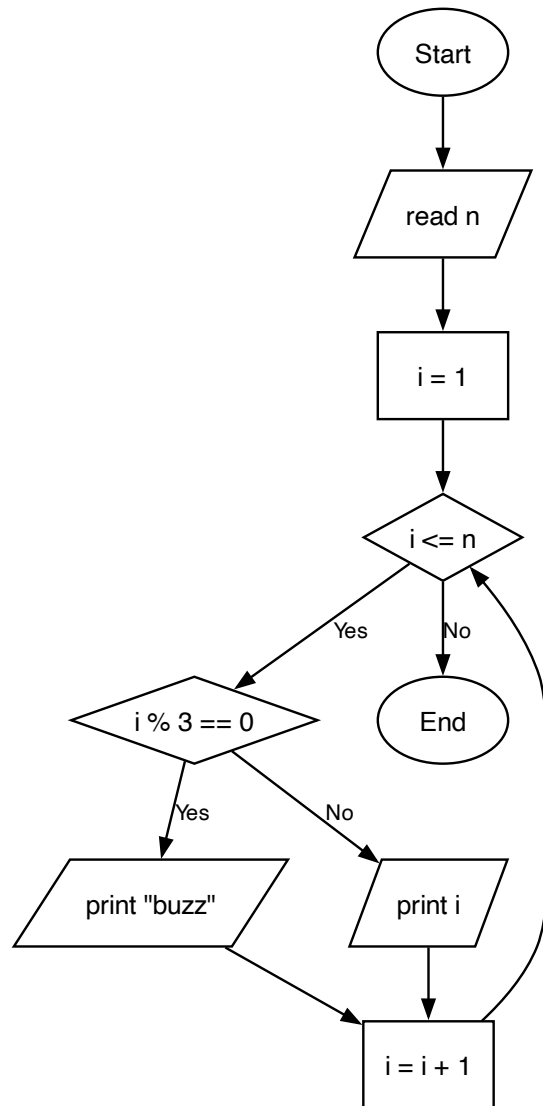
*Note:  $n // 10$  means whole-number division — divide and drop the remainder. For example,  $472 // 10 = 47$  (not  $47.2$ ).*



| n   | count | output |
|-----|-------|--------|
| 472 |       |        |
| 472 | 0     |        |
| 472 | 1     |        |
| 47  | 1     |        |
| 47  | 2     |        |
| 4   | 2     |        |
| 4   | 3     |        |
| 0   | 3     |        |
| 0   | 3     | 3      |

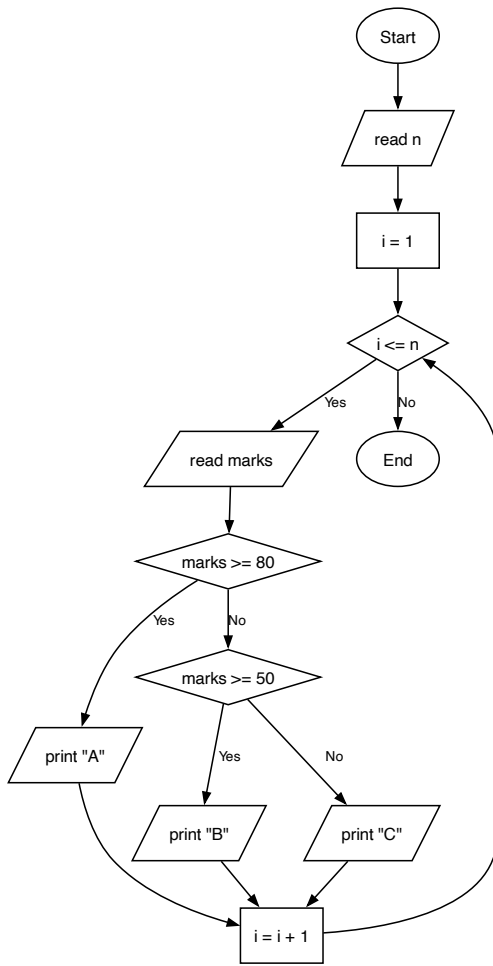
**Output:** 3

## Problem 6.



### Problem 7.

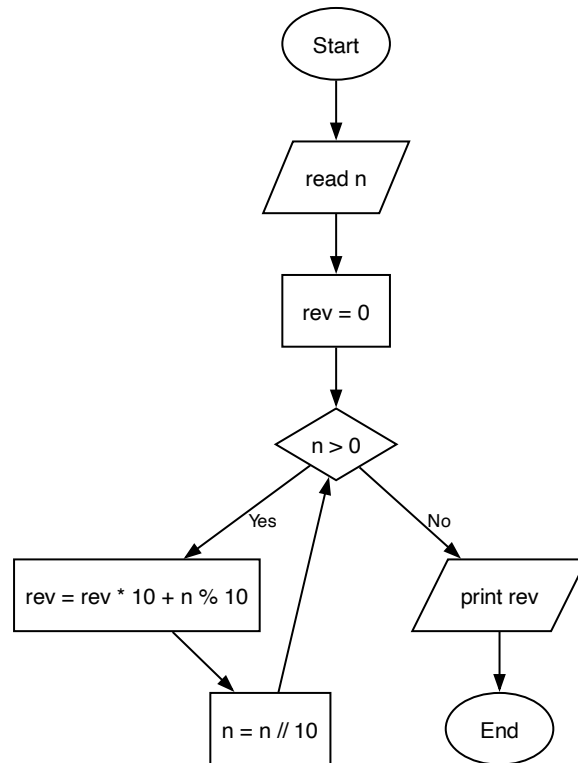
Trace this algorithm for  $n = 3$ , where the values of *marks* entered are 55, 82, 40 (in that order), and complete the trace table.



| n | i | marks | output |
|---|---|-------|--------|
| 3 |   |       |        |
| 3 | 1 |       |        |
| 3 | 1 | 55    |        |
| 3 | 1 | 55    | B      |
| 3 | 2 | 55    |        |
| 3 | 2 | 82    |        |
| 3 | 2 | 82    | A      |
| 3 | 3 | 82    |        |
| 3 | 3 | 40    |        |
| 3 | 3 | 40    | C      |
| 3 | 4 | 40    |        |

**Output:** B, A, C

### Problem 8.



(i) For each input below, follow the flowchart and write the output.

*Note:  $n \% 10$  gives the remainder after dividing by 10 — the last digit of  $n$ . For example,  $472 \% 10 = 2$ .*

| input $n$ | output |
|-----------|--------|
| 123       | 321    |
| 47        | 74     |
| 5         | 5      |
| 470       | 74     |

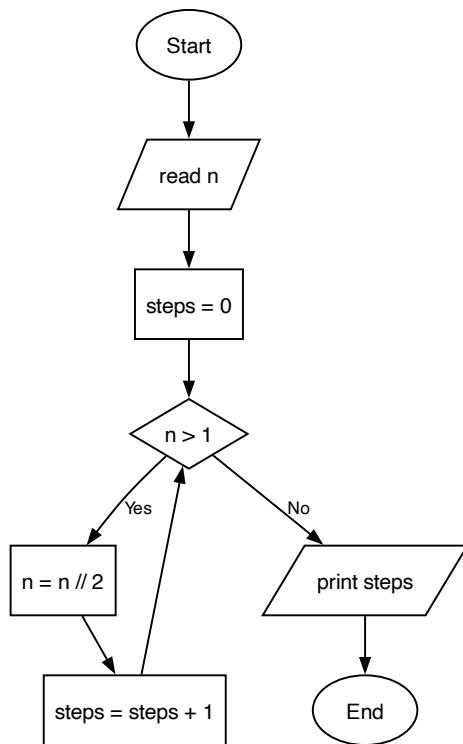
(ii) Can you tell what this algorithm is doing?

**Answer:** Reverse the digits

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### Problem 9.

(i) Trace this algorithm for  $n = 16$ , and complete the trace table.



| n  | steps | output |
|----|-------|--------|
| 16 |       |        |
| 16 | 0     |        |
| 8  | 0     |        |
| 8  | 1     |        |
| 4  | 1     |        |
| 4  | 2     |        |
| 2  | 2     |        |
| 2  | 3     |        |
| 1  | 3     |        |
| 1  | 4     |        |
| 1  | 4     | 4      |

(ii) For each input below, follow the flowchart and write the output.

| input $n$ | output |
|-----------|--------|
| 8         | 3      |
| 16        | 4      |
| 100       | 6      |
| 1000      | 9      |

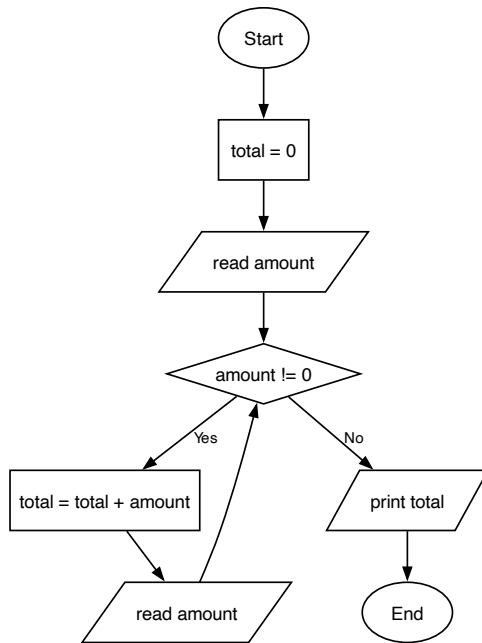
(iii)  $n$  grew from 8 all the way to 1000 — but look how little the output grew! Can you explain why?

**Answer:** Counting halvings — doubling  $n$  adds only ONE step. This is the secret behind binary search!

### Problem 10.

*At the end of the day, a shopkeeper adds up the money received from each customer. She enters the amounts one by one, and enters 0 when there are no more. The program then shows the total. Notice that the algorithm cannot know in advance how many customers there were!*

Trace this algorithm, where the values of *amount* entered are 120, 75, 200, 0 (in that order), and complete the trace table.



| total | amount | output |
|-------|--------|--------|
| 0     |        |        |
| 0     | 120    |        |
| 120   | 120    |        |
| 120   | 75     |        |
| 195   | 75     |        |
| 195   | 200    |        |
| 395   | 200    |        |
| 395   | 0      |        |
| 395   | 0      | 395    |

**Output:** 395